

Experiment 20: Create a GitHub Repository and Implement Version Control

Aim

To create a GitHub repository, implement version control, manage branches, merge changes, resolve conflicts, and document the workflow.

Tools Required

- Git
- GitHub
- VS Code (or any text editor)

Procedure

Step 1: Create a Repository

- Create a new GitHub repository with a README file.

Step 2: Clone the Repository

Bash

```
git clone <repository-url>  
cd <repository-name>
```

Screenshot: Repository cloned.

```
srinidhi@Srinidhis-Laptop ~ % cd desktop  
srinidhi@Srinidhis-Laptop desktop % git clone https://github.com/Sriarevalagan23/Test-repo-fork  
Cloning into 'Test-repo-fork'...  
remote: Enumerating objects: 79, done.  
remote: Counting objects: 100% (79/79), done.  
remote: Compressing objects: 100% (68/68), done.  
remote: Total 79 (delta 5), reused 76 (delta 5), pack-reused 0 (from 0)  
Receiving objects: 100% (79/79), 1.41 MiB | 132.00 KiB/s, done.  
Resolving deltas: 100% (5/5), done.  
srinidhi@Srinidhis-Laptop desktop % cd test-repo-fork  
srinidhi@Srinidhis-Laptop test-repo-fork %
```

Step 3: Create Branches

Bash

```
git checkout -b module1  
git checkout -b module2
```

Screenshot: Branches created.

Step 4: Make Changes

- Modify or add a file (e.g., README.md or a sample file).

Step 5: Commit and Push

Bash

```
git add .
```

```
git commit -m "Added module changes"
```

```
git push origin module1
```

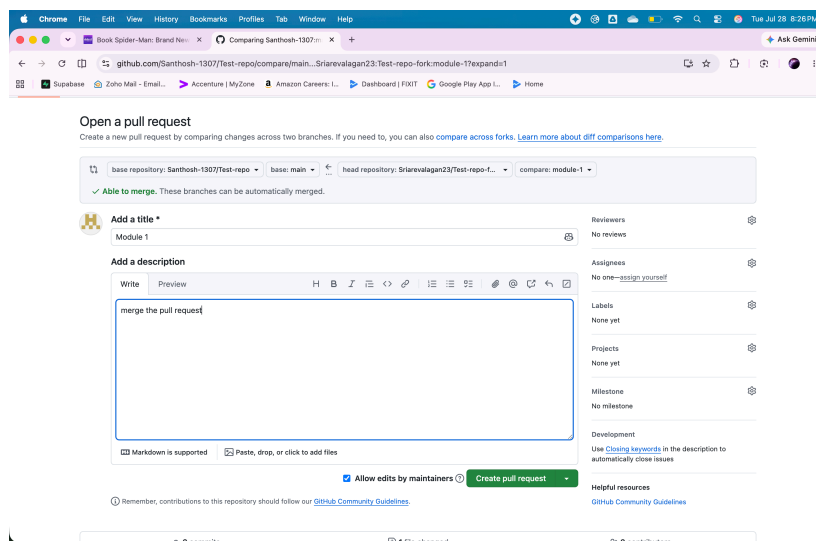
Screenshot: Changes pushed.

```
srinidhi@Srinidhis-Laptop Test-repo-fork % git push origin module-1
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 307 bytes | 307.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'module-1' on GitHub by visiting:
remote:   https://github.com/Sriarevalagan23/Test-repo-fork/pull/new/module-1
remote:
To https://github.com/Sriarevalagan23/Test-repo-fork
 * [new branch]      module-1 -> module-1
srinidhi@Srinidhis-Laptop Test-repo-fork %
```

Step 6: Create a Pull Request

- Create a Pull Request on GitHub for the feature branch.

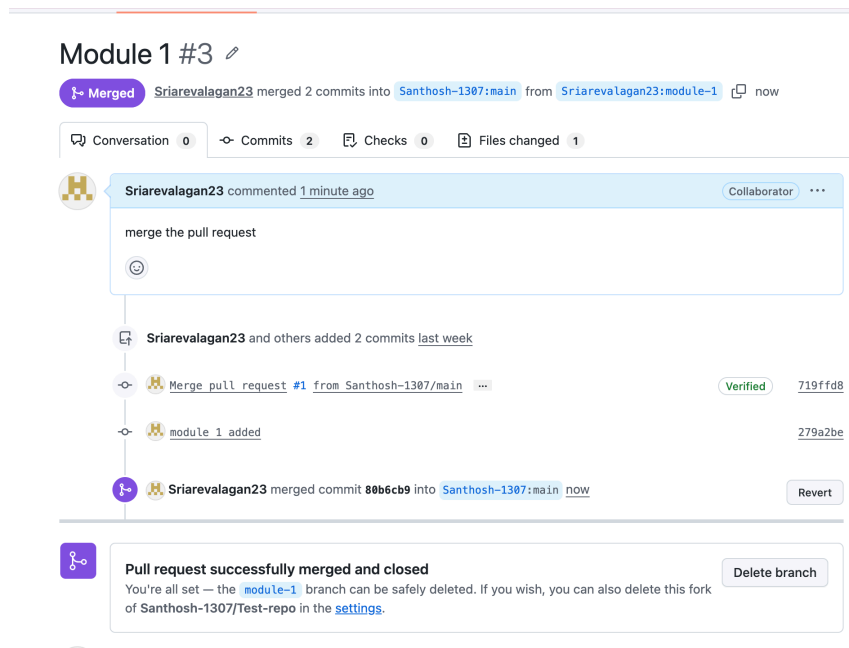
Screenshot: Pull Request created.



Step 7: Merge and Resolve Conflicts

- Review and merge the Pull Request.
- Resolve merge conflicts if they occur.

Screenshot: Merge completed.



Step 8: Update README

- Document the project workflow in the README file.

Screenshot: Updated README.

Step 9: Submit Repository

- Submit the GitHub repository link.

Link: <https://github.com/Santhosh-1307/Test-repo>

Result

Successfully created a GitHub repository, managed branches, merged changes using Pull Requests, resolved conflicts, and documented the workflow.